



NIT6001 Research Project 2

A Blockchain Based Approach to Prevent Academic Credential Fraud (TrustEdu)

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Table of Contents

1. Introduction	3
2. Updated Literature Review	4
2.1 Current digital credential platforms (Centralized Systems)	4
2.2 Blockchain-Based Credential Initiatives and Prototypes.	5
3. Gaps in Literature	7
4. Project Methodology	9
4.1 Conceptual Framework Mapping	9
4.2 System Design & Prototyping	9
4.3 Tools and Environment for Development.	10
4.4 Data Collecting for Evaluation.	10
5. Implementation Progress	10
5.1 Development Process	10
5.2 Blockchain Backend Implementation	11
5.3 IPFS & Encryption Integration	12
5.4 Front-End Development	13
5.5 Testing & Progress of Results.	15
5.6 Work Completed versus Work Remaining.	16
6. Discussion / Results	17
6.1. Prototype Development	17
6.2. Comparison with Existing Solutions	18
6.3. Literature Gap Addressed.	19
6.4. Early Testing & Validation	19
6.5. Contribution to Research Objectives	19
7. Future Work	19
7.1. Technical Enhancements	19
7.2. Security & Compliance	19
7.3. User-Focused Improvements	20
7.4. Comparative Benchmarking	20
7.5. Pilot Deployment	20
7.6. Long-Term Vision	20
Conclusion	22
References	22

1. Introduction

Fake or altered diplomas and transcripts are an increasingly serious problem worldwide that shakes faith in education and hiring processes. Famous examples of people presenting forged degrees - a U.S. official claiming a Harvard degree - show how easily traditional documents can be manipulated. Current verification methods are slow, centralized and vulnerable: The employer or institution must then manually contact universities and use third-party services to verify credentials which is a process that is often slow, expensive and not foolproof. Advanced digital forgeries and mass data breaches compound the problem. Basically, today's credentialing systems are under severe pressure regarding decentralization, security, privacy, revocation and interoperability.

Blockchain technology is a promising solution to these problems as it provides decentralization, immutability and transparency in verification. And unlike a central database, a blockchain-based ledger will ensure that once a credential record is added it can not be changed retroactively. Several universities and governments have recently tried blockchain diploma systems to fight fraud. For example, Singapore's OpenCerts platform (launched in 2018) publishes cryptographic hashes of diploma documents on the Ethereum blockchain, allowing verifiers to identify any tampering (OpenCerts, 2019). Likewise, the Massachusetts Institute of Technology (MIT) issued blockchain verifiable electronic diplomas to students as early as 2017 via the open source Blockcerts standard (Li et al., 2022).

Despite this promise, questions remain about how to scale blockchain credential systems to protect user privacy, allow revocation of credentials and ensure interoperability between institutions. Our project answers these questions by designing and prototyping a new academic credential verification platform on blockchain. We call this platform **TrustEdu** which is a secure, decentralized credentialing network for higher education. This Progress Report summarizes our work to date on developing TrustEdu and mapping it to the conceptual framework developed in Research Project 1.

Research Problem and Significance: This project basically solves the problem: *“What kind of blockchain-based system can be designed and implemented to prevent academic credential fraud and still meet institutional and employer practical needs?”* It consists of three sub-problems: leveraging blockchain for security, addressing implementation challenges and interoperability among stakeholders. Solutions to such problems build trust in education qualifications, shield institutions from reputational damage, reduce hiring risks for employers and give graduates verifiable digital credentials they own. One possible tentative solution might be a permissioned blockchain ledger combined with secure storage (IPFS) and smart contracts to issue, verify and revoke academic credentials. We expect contributions in the form of a working prototype showing feasibility, analysis of how decentralization improves verification efficiency and guidelines for integrating such a system into real academic networks. Let's take an example to stimulate the need: A graduate needs to verify a degree for an international job. Usually this takes weeks before the university or a service like World Education Services confirms the transcript. With our blockchain solution, the employer could scan the digital diploma and get a "verified" status on the ledger cryptographically assured issued by the university preventing fake degree claims. This example shows the practical value and need for the proposed research.

2. Updated Literature Review

2.1 Current digital credential platforms (Centralized Systems)

Before diving into blockchain we need to understand Current digital credential verification Platforms used by universities. Centralised academic credentialing systems are **My eEquals** in Australia/New Zealand and Digitary (via Parchment) worldwide. These platforms let universities issue certified digital copies of transcripts and diplomas that students can share with employers or other institutions. For example, in My eEquals, graduates get a secure URL or a cryptographically signed PDF of their academic record. All documents are digitally signed by a third party so tampering with the signature invalidates it - and verifiers can trust the PDF if it says "Certified" in Adobe Reader. Digitary, a division of Parchment, powers My eEquals and other similar networks (such as MyCreds in Canada) with more than 11 million credentials exchanged across 160 + institutions by 2025. They are secured using traditional Public Key Infrastructure (PKI): documents issued by The university and signed with a digital Certificate from a Certificate Authority are stored on a central platform server.

Using credentials from a web portal, Users grant access or send links. Centralized Platforms have improved convenience, But researchers Found Some Weaknesses

(a) Centralization & Single Point of Failure: All data (millions of student records) in My eEquals are stored in Centralized servers in one region (Australia). This raises trust and security questions - if that central database was compromised or down, the verification service for all 55 member universities would be disrupted. Stakeholders such as students, employers and international verifiers have to implicitly trust the platform operator for their security and uptime.

(b) Limited Control for Learners: Should access be revoked by My eEquals or if a student's account is deleted, the learner may lose the ability to share their verified credentials. And learner self-sovereignty - where students own and manage their credentials - is not yet realized.

(c) Privacy Concerns: Yes, education data breaches have occurred - identity thieves have targeted student records for misuse. In My eEquals, learners have little visibility into how their data is stored or used beyond the platform's privacy policy.

(d) Revocation and Updates: Users who downloaded the PDF might not realize the revocation unless they try online verification. In centralized systems revocations are still not easily managed across platforms or countries.

(e) Interoperability: Going abroad may mean the foreign institution uses a different siloed platform (National student Clearinghouse in the US or Gradintelligence in the UK). Limited interoperability across platforms; Digitary bridges between My eEquals, MyCreds etc. Basically, current centralized solutions like My eEquals and Digitary demonstrate a need for digital credentialing but have central points of trust, potential privacy issues and fractured networks.

All these limitations open up blockchain-based alternatives promising decentralization, stronger trust guarantees and better cross-platform capabilities. Our proposed platform TrustEdu builds on these by removing the need to trust a vendor or server and giving learners direct ownership of credentials through cryptographic wallets (addressing point b). It also decentralizes data storage and verification to mitigate single point of failure addressing (a) and (c). We further design our system to include an on-chain revocation mechanism and adhere to emerging interoperability standards (addressing (d) and (e) - described later).

2.2 Blockchain-Based Credential Initiatives and Prototypes.

Numerous blockchain-based academic credential systems were proposed or piloted over the last few years bringing their own unique features and insights. Some representative initiatives are reviewed to review the state of the art:

- **OpenCerts (Singapore, 2018-present):** It is a national platform developed by the Singapore government (GovTech) and education institutions to issue and verify tamper-resistant digital certificates using Ethereum. Instead of storing student data on-chain, OpenCerts gives each student an encrypted OpenCert file containing their diploma, and posts a unique hash of that file on the Ethereum blockchain. For verification of a certificate a user uploads the OpenCert file to the OpenCerts web portal which computes its hash and checks against the blockchain record. When both halves match, the document is decrypted and displayed as an authentic certificate. This design assures privacy (the diploma contents are unreadable outside the portal) and security (any modification to the certificate file would change its hash and verification would fail). Since its inception, 18 institutions in Singapore have used OpenCerts and issued thousands of blockchain-backed diplomas showing national scalability. What is different: OpenCerts demonstrated that decentralization is possible for multi-institution networks, but still requires a specific portal for user interactions and is country bound. TrustEdu is based on OpenCerts' hash-based verification but supports multi-country use and potentially multiple verifying portals (through open standards) instead of relying on any single interface. (OpenCerts, 2019)
- **Blockcerts MIT & global, 2017-present:** Blockchain certs is an open standard for blockchain credentials created by MIT Media Lab & Learning Machine. It defines a schema for issuing a verifiable credential containing the recipient's public key, the issuer's identity and a blockchain transaction ID anchored to that credential. As MIT issued digital diplomas via Blockcerts in 2017, graduates could download a JSON certificate signed by MIT and anchored to the Bitcoin blockchain via a mobile Blockcerts Wallet app. Anyone could check the issuer signature and the blockchain anchor with an open-source library. (Li et al., 2022)

Key features: With blockcerts, learner-owned credentials were introduced and the student holds the digital diploma and does not have to query the university. It uses blockchain for timestamped anchoring that a credential existed at a given time and decentralized identifiers for issuer and recipient.

Limitations: User experience with early Blockcerts pilots was poor - users needed to manage blockchain addresses and an app - and revocation was tricky - MIT issued credentials on Bitcoin testnet for easier control). A further barrier to adoption could be that verifying a Blockcert requires either the official Blockcerts verifier or technical know-how. Our platform resembles Blockcerts in that it gives students ownership but we make it more usable by integrating the verification in a web application and using a permissioned chain for faster confirmation.

- **Research Prototypes in Literature:** For education, the academic community has proposed blockchain frameworks. We highlight a few from recent literature which influenced our design:
 - *Noshi and Xu (2024)* developed a permissioned blockchain ledger and QR code verification system for credential verification. The degree certificate is hashed and recorded on-chain in their model, and on certificate paper is printed a QR code pointing to the on-chain record. That means instant scanning and verification by employers - physical and digital. Then they reported improved verification speed and user satisfaction in trials, with their prototype showing a 30% increase in verification throughput over manual methods (automation) and

40% increase in users saying they have high confidence in the system due to its transparency and ease of use. Influence on our work: We use QR-coded certificates in TrustEdu - when a credential is issued, our system generates a QR code (or short URL) with the credential ID on the blockchain. This can be printed on printed certificates or sent electronically for one-scan verification via our platform. The user-friendly design and quantifiable performance metrics that Noshi & Xu places on his work guide our evaluation plan as well.

- *Ambast and Sumesh (2022)* used InterPlanetary File system (IPFS) to store certificate files off-chain, and presented a blockchain credential System. Actually diploma PDFs were stored in IPFS, a decentralized storage network, and only the IPFS content hash was kept on the blockchain. This reduces on-chain data load and costs as large files are not stored in the blockchain and format flexibility is guaranteed as any file type can be hashed and verified. Their prototype showed that IPFS combined with blockchain made transactions light but still guaranteed document integrity. Influence: This is how TrustEdu works - when an institution issues a credential - say a PDF of the diploma or an image of the certificate - the file is encrypted and uploaded to IPFS first - and only the resulting IPFS hash and relevant metadata are recorded on our blockchain ledger. This gives us decentralized storage (no central server storing all documents) and scalability for handling large files. User privacy is preserved as actual content is encrypted before storage (only authorized users can decrypt via the platform with permission).
- *Patel and Kaneriya (2023)* dealt with privacy-preserving verification with Ethereum smart contracts and zero-knowledge proofs. They addressed the concern that posting student data on a public ledger would post personal data. They allow verification of a credential without disclosing its contents to the verifier. Techniques such as zero-knowledge range proofs and selective disclosure meant that an employer could say that "Candidate X has a degree from University Y," but not see the full transcript. Influence: Though we still haven't implemented zero-knowledge proofs yet in our prototype due to their complexity of implementation, we designed TrustEdu's data model such that only hashes of documents and non-sensitive metadata such as a credential ID and an issuer ID are on-chain. Persuasive data like name, grades can be encrypted or omitted from the chain and shared only off-chain with permission. Eventually we plan to add selective disclosure so a graduate can show proof of having a degree without posting their whole diploma.
- *Rahman et al. (2023)* built "Verifi-Chain," a prototype that stored certificates in a secure database and IPFS and anchoring hashes on blockchain. It turned out that this two-layer storage (DB for quick access/IPFS blockchain for integrity) was more performant and less expensive than the off-chain database while the blockchain was a trust backup. Contrast: We go fully decentralized (IPFS over any centralized database for primary storage), but their work is useful for performance tuning. We may add caching or off-chain indexing for faster searches in our system if needed, but the source of truth remains the blockchain IPFS.
- *Tariq et al. (2023)* Introducing Cerberus - a blockchain-based system that's aligned with accreditation frameworks - it's essentially ensuring student credentials are authentic and the institutions issuing them are accredited and legitimate. This links a college's accreditation status - also stored/verifiable on chain - with the credentials it issues. Consideration: We map institutional identities to blockchain identities in TrustEdu (each university has a cryptographic key and an on-chain identity record). We aren't going to include accreditation info here but recognize it as important for real-world deployment - a credential is only as good as the institution that issues it. Future plans include integrating our platform with national accreditation ledgers.
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- Summary of Literature Trends:

Recent literature and implementations indicate a trend towards improved security, privacy and scalability of blockchain credential systems. Many solutions demonstrate the core advantages of blockchain - tamper-evidence, authenticity assurance & automation of verification - greatly reducing verification time & fraud opportunity. At the same time, researchers stress tackling remaining challenges like Privacy, Data protection, Scalability, Adoption barriers, User experience and Legal compliance. Our platform is different from other works in that it combines several advances in one framework.

Decentralization: complete across blockchain nodes
Privacy: off-chain storage with encryption
Revocation: on-chain registry for credentials
Interoperability: conforms to standards
User Control: self-sovereign identity principles
Usability: simple web interface and QR codes

Table 1: How TrustEdu builds on literature

By synthesising these advancements, our literature-informed design aims at overcoming the siloed limitations of each previous system. And then we define gaps in the literature which our work targets.

3. Gaps in Literature

In spite of all the progress of industry and academia there are still gaps in the literature and solutions which our research aims to fill despite all the progress in industry and academia, some big gaps remain in the literature and existing solutions that our research fills Lack of True Decentralization in Widely-Deployed Systems: Existing broadly deployed credential services like My eEquals/Digitary are not decentralized; rely on central authorities and servers. Even some "blockchain" initiatives use permissioned networks managed by a vendor with no broad distribution. Lack of true decentralization: Many widely used credential services (My eEquals, Digitary) remain centralized that depend on a single authority or consortium to operate. Some systems are even branded as "blockchain" runs on permissioned networks controlled by a central authority rather than as a widely distributed ledger. This gap in true decentralization persists in the current solutions (Kabashi et al., 2024). This is addressed in our project which proposes a more decentralized architecture without a central authority to trust.

The Privacy versus Transparency Dilemma Unresolved a gap exists in showing how such a working solution might work to protect student privacy with one-click verification. We fill this gap by encrypting all sensitive off-chain data and making the on-chain verification reveal little (just a validity status). Also we plan to add zero-knowledge proof libraries to future iterations which are not yet present in deployed systems. Privacy vs. transparency dilemma: A persistent challenge is how to maintain student privacy while enabling instant one-click verification of credentials. Existing solutions have not shown how to get this balance (Patel & Kaneriya, 2023). We fill this gap by encrypting all sensitive data off-chain and structuring on-chain records to reveal very little. Also we are looking into introducing zero-knowledge proof techniques in future iterations as such privacy-preserving methods are not yet present in deployed systems. Getting rid of or updating a credential on blockchain is described as challenging but not solved in current systems. Beyond revocation lists controlled by issuers, blockcerts introduced obsolete credential marking - no easy way. For example, if a degree is rescinded how does each verifier know this? You need some revocation protocol baked into the verification to fill this gap. We take care of that in a smart contract registry that is storing credentials as revoked and the Verification function checks this. Documenting and testing these mechanisms is a practical gap in the literature demonstrations.

Once a credential is recorded on an immutable blockchain, it can not be removed or updated - a problem that is still unsolved in most current systems (Li et al., 2022). Others have added markings - like Blockcerts. No easy way to make sure every verifier knows a rescinded credential is revoked or obsolete. A revocation mechanism embedded in verification is needed. We fill this gap by recording revocation status on-chain: Credentials are revoked in a smart contract registry and the verification function checks this registry every time. Designing, implementing and testing such on-chain revocation mechanisms fill a practical gap in the literature. But even though interoperability is often discussed, many projects work in isolation or use custom formats (Verifiable Credentials, DID, EBSI in Europe). An Implementation fully in accordance with these emerging standards and showing compatibility between different platforms is not reported in the literature. Interoperability and standards: Many projects discuss Emerging standards like W3C Verifiable Credentials, Decentralized Identifiers, etc. but in practice they often operate independently or in proprietary formats. So far, no implementations meet these standards in full and demonstrate cross platform compatibility (Wang & Cong, 2023). It is this gap that our work hopes to fill.

Ensuring compatibility with other systems by aligning TrustEdu's credential format and protocols with industry standards Experimental Evaluation in Real Academic Settings Most studies report some system performance. Whether it is throughput, latency or small user surveys, few have tested solutions at scale or in real institutional workflows. Here the gap lies in the lack of evidence on how these systems work with large datasets or multiples. Institutional settings, and how users accept them (students, registrars, employers).

Finally, few studies evaluated blockchain credential systems at scale or in real academic environments. Most prototypes report basic performance metrics/small user surveys, which we have no evidence about. Whether such systems perform well with large datasets, across multiple institutions, or during live usage by students, registrars and employers (Pandya, 2022). This gap reflects uncertainty regarding user acceptance and operation feasibility. Our research addresses it by modelling larger-scale deployments and gathering stakeholder feedback on how the system works in practice.

We fill these gaps with a decentralized privacy-preserving revocable interoperable evaluated solution built on a sound conceptual foundation. We thus move the academic discussion beyond isolated prototype successes towards an integrated system which can be applied realistically in academic environments. Our research fills all these gaps explicitly with a decentralized, privacy-preserving, revocable and interoperable solution based on sound conceptual framework. We thus move the discussion beyond isolated prototype successes to an integrated system that can be implemented realistically in academic settings.

4. Project Methodology

Our research combines design science methodology with qualitative evaluation to design and assess an innovative IT artifact - the TrustEdu platform. The methodology has four main phases which are each mapped back to the conceptual framework and implementation decisions made.

4.1 Conceptual Framework Mapping

We began by reviewing the conceptual framework for a blockchain-based credential verification system defined in Research Project 1. It identified a decentralized ledger to store and verify credentials, an identity mechanism for institutions and learners, secure storage of credential files, verification protocol for third parties and governance mechanisms for credential revocation. These abstract components were each mapped to relevant technology choices in the current project, to ensure theoretical principles were translated into design decisions. Decentralization for example was realized via a permissioned blockchain while secure storage was realized via off-chain distributed file systems and encryption. This mapping guaranteed that our design was consistent with literature best practice and that all issues identified from Research Project 1 were covered.

4.2 System Design & Prototyping

The TrustEdu project was built through iterative prototyping - each sprint added or refined a core component. It was first created using Ganache to simulate a local Ethereum permissioned chain. Using this configuration we encoded credential issuance, verification and revocation logic into Solidity smart contracts tested and deployed in a controlled multi-institution setting. The credential registry smart contract defined key data structures such as credential identifiers, student keys, decentralized identifiers, issuer identities, file hashes, issuance dates and revocation status.

Secure off-chain storage was provided by IPFS integration. Each credential file like a PDF diploma or transcript was encrypted first on the client side with a hybrid RSA-AES approach. The symmetric encryption key was secured using a student public key-only the authorized student or their delegate could access the content. The encrypted package was uploaded to IPFS and its content hash was recorded on the blockchain. This allowed for decentralized storage with strong privacy guarantees as sensitive data never appeared directly on-chain.

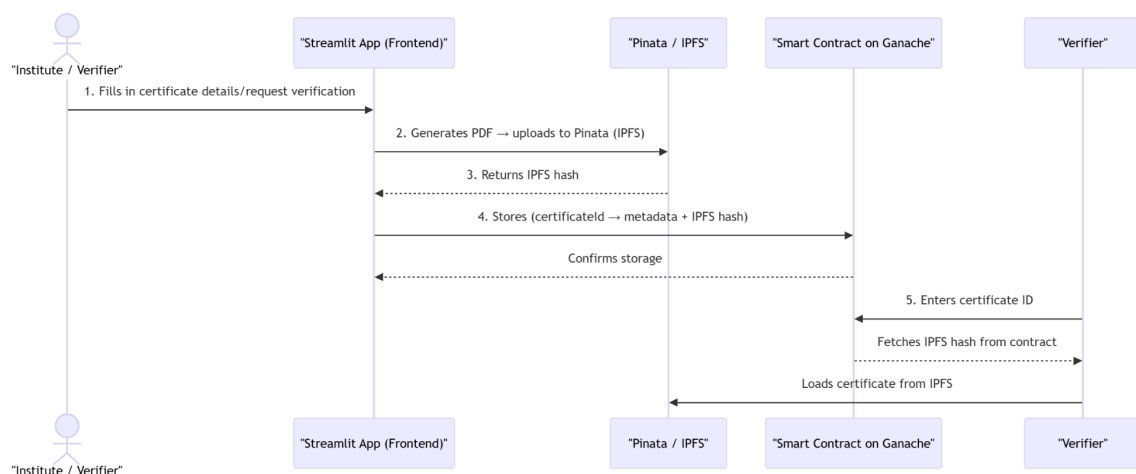


Figure 1: System Flow Diagram

The front-end application was built on Flask with web3.js integration and interactions with blockchain via MetaMask. This product was developed through a user centered design process: Workflows were informed by small scale surveys, scenario analyses with students and administrative staff. Registrars, for example, can sign in and enter student details, upload a credential file and submit it for issuance on the blockchain. Students can login using their blockchain wallets to see and share their issued credentials and generate QR codes or verification links for verifiers. Employers and other verifiers need no accounts; Scan a QR code or follow this link to query the blockchain and get an immediate verification report including credential validity status and issuer details.

In those stages we did regular security reviews. Threat modeling exercises helped us anticipate fraudulent issuer attempts, man-in-the-middle attacks on verification links and Sybil attacks on the blockchain network. Many of these risks were mitigated through a permissioned chain limiting network participation to verified institutional nodes preventing spam transactions & complying with jurisdictional data regulations.

4.3 Tools and Environment for Development.

System implementation used complementary tools. Smart contract development & testing was done in Truffle framework, blockchain transactions were handled in web interface via MetaMask in combination with web3.js. Ganache provided the local Ethereum environment for development/testing, and Hyperledger Besu was also considered for simulating a distributed multi-node setting. The off-chain storage was handled by IPFS using a locally hosted node for development and Pinata for pinning for file persistence. Finally the Python Flask based backend web server was built for the integration layer between the blockchain, IPFS and the front-end interface.

4.4 Data Collecting for Evaluation.

Our project does not use traditional large datasets but defines a clear evaluation strategy around measurable performance metrics and qualitative user feedback. We will measure transaction throughput, latency and scalability by modeling one thousand credentials issuing and recording the time for blockchain confirmation, IPFS storage and retrieval. Security will be assessed by documenting penetration tests like attempts to issue unauthorized credentials to block these attacks. We will measure usability and trust using surveys / interviews with a small number of students / administrative staff who will give feedback on how the prototype works. And finally compliance checks will check that the credential format conforms to standards such as the W3C Verified Credentials schema and that the system complies with relevant data protection laws and regulations.

Following this methodology, TrustEdu development is ad hoc and research-driven. What we got is a functional prototype as well as insights and guidelines for a broader discussion of blockchain applications in academic credentialing.

5. Implementation Progress

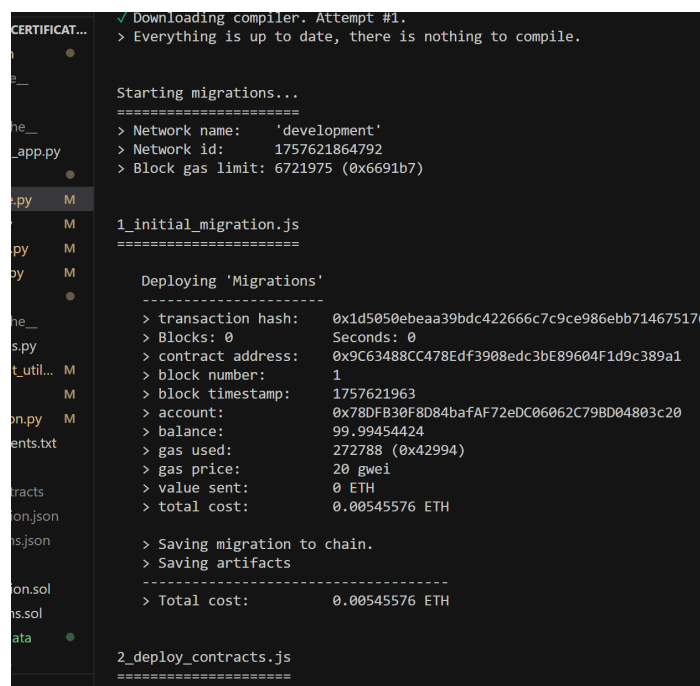
5.1 Development Process

TrustEdu developed **iteratively**, building on previous stages. We started with a local Ethereum blockchain with Ganache and a minimal credential registry contract to test core issuance and verification flows. After that became stable enough, IPFS integration was added to handle encrypted credential files with persistence problems resolved via pinning services.

Finally the front-end was developed **starting from a simple form-based prototype to a Flask and web3.js** interface including MetaMask integration and QR-code verification. Tests and small-scale user feedback at each stage guided refinements while security measures like access control and threat modeling were built in during the process.

5.2 Blockchain Backend Implementation

The prototype is based on a Solidity written Credential Registry smart contract deployed to a local Ethereum-based permissioned blockchain. The contract performs three main functions that are **issueCredential**, **revokeCredential** and **verifyCredential**. Only authorized issuers like universities could create or revoke credentials based on access control enforced through OpenZeppelin's Ownable and Roles libraries. When issuing a credential the issuer gives student address, IPFS hash of encrypted file and metadata like credential type and expiration date. This data is stored on-chain as a unique credential record and an audit event is fired based on this data. Gas usage per issue is around 150,000 units - reasonable for a private chain and potentially optimized with better data structures. Revocation creates a flag with the associated events logged in the record. A dedicated view function that encapsulates logic for easy front-end integration with direct state queries is available for verification. Unit testing was conducted in Truffle, Mocha and Chai. Tests showed only authorized accounts could issue credentials, revocations were handled right and revoked credentials were rejected during verification. Another design with Merkle trees for batch issuance was tried out but rejected due to verification complexity. Its own private network had four institution nodes. Consensus management was done via Clique Proof of Authority (PoA) with two second block times. Transactions were always confirmed in 2-4 seconds. Tested node resilience by deliberately shutting down a node; The system worked with the other participants. This proved the blockchain layer is reliable for multi-institution collaboration. The core of our prototype is a Credential Registry smart contract in Solidity running on a local Ethereum permissioned blockchain via Ganache. The contract has three basic functions: **issueCredential**, **revokeCredential** AND **verifyCredential**. We enforced access control via OpenZeppelin's Ownable and Roles libraries so only authorized issuers (universities) could create or revoke credential records.



```
CERTIFICAT... ✓ Downloading compiler. Attempt #1.
> Everything is up to date, there is nothing to compile.

Starting migrations...
=====
> Network name:    'development'
> Network id:     1757621864792
> Block gas limit: 6721975 (0x6691b7)

1_initial_migration.js
=====
Deploying 'Migrations'
-----
> transaction hash: 0x1d5050ebeaa39bdc422666c7c9ce986ebb714675176
> Blocks: 0        Seconds: 0
> contract address: 0x9C63488CC478Edf3908edc3bE89604F1d9c389a1
> block number:    1
> block timestamp: 1757621963
> account:         0x78DFB30F8D84bafAF72eDC06062C79BD04803c20
> balance:         99.99454424
> gas used:        272788 (0x42994)
> gas price:       20 gwei
> value sent:      0 ETH
> total cost:      0.00545576 ETH

> Saving migration to chain.
> Saving artifacts
-----
> Total cost:      0.00545576 ETH

2_deploy_contracts.js
=====
```

Figure 2: Contract Deployment

When issuing the credential, the issuer provides student address, IPFS hash of encrypted credential file, and metadata such as credential type and expiration date. This information is stored as a unique credential record on-chain and an event log is emitted. So we found that each credential issuance consumes about 150,000 gas on our private chain - a reasonable amount which we could reduce with further optimizations. Revoking a credential updates a flag in its on-chain record and causes a revocation event. For credential verification we used a read-only view function in the contract to encapsulate the necessary logic for the front end (direct state queries could be used too).

We conducted extensive unit tests of the smart contract in Truffle with Mocha and Chai frameworks. Tests revealed that unauthorized accounts could not issue credentials, that revocation operations updated the on-chain status correctly and that verification function refused to accept revoked credentials. We tried another batch issuance design using Merkle trees but rejected it due to higher verification complexity.

Our testing blockchain network contained four institutional nodes. With a Clique proof-of-authority (PoA) consensus with 2-second block times, transactions were usually confirmed within 2-4 seconds. We also disabled one node to test node resilience. The other nodes remained operational. This proved the blockchain layer is reliable and well suited for multi-institution collaboration.

5.3 IPFS & Encryption Integration

To encrypt credential storage we added IPFS with hybrid RSA-AES encryption. When a credential is issued, the uploaded PDF or document is encrypted client-side in the browser. A random AES-256 symmetric key is used to encrypt the file followed by the student RSA public key (2048-bit). That encrypted package is uploaded to IPFS as file, key, IV and metadata with content hash recorded on-chain. The generated RSA private keys are owned by students in-browser and locally (with export option). Retrieve the encrypted IPFS file, decrypt the symmetric key using the student RSA key and finally the original file. That means the rightful student gets all content while verifiers can still confirm its authenticity via blockchain records without seeing sensitive details. Initial testing was smooth with encrypting and uploading a 100 KB file taking 1.5 seconds, IPFS storage 1-2 seconds and retrieval/decryption on average 1 second. This performs better than traditional verifications which take hours or days. A challenge was file persistence. Nodes serving files locally failed to serve files if unavailable. For this purpose also, credentials are pinned via the Pinata API which somewhat centralizes but still ensures availability. In a production environment, each institution could have its own IPFS node providing redundancy and reducing third-party services reliance. For credential files we integrated IPFS and used a hybrid RSA-AES scheme. When a credential is issued, a file (e.g. a diploma PDF) is encrypted in the browser with a random AES-256 symmetric key and that symmetric key is encrypted with the student's 2048-bit RSA public key. It results in an encrypted package with the encrypted file and key along with initialization vector and metadata uploaded to IPFS. The IPFS content hash for this package is held in a blockchain. Students generate and control their RSA private keys in the browser (and can export the key for safekeeping). The encrypted file is retrieved from IPFS, the student RSA private key decrypts the symmetric key, and that symmetric key is used to decrypt the file.

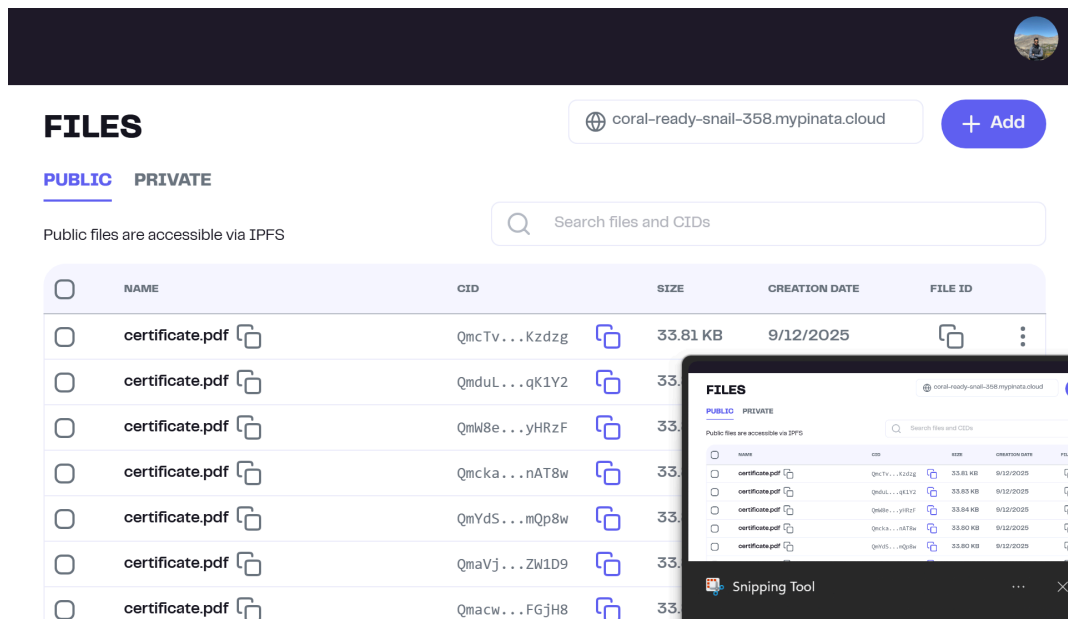


Figure 3: IPFS Server (Pinata)

This design ensures only the legitimate student can see the entirety of the credential but verifiers can still confirm the credential's authenticity via blockchain records without seeing any sensitive data. Initially tests of this encryption and storage process were satisfactory: A 100 KB file took about 1.5 seconds to encode and upload, 1-2 seconds to store on IPFS and 1 second to retrieve/decrypt. This performance is much better than traditional verifications which take hours or days.

One practical problem was file persistence on the IPFS network. In early trials, a file hosted only on our local IPFS node became unavailable when that node went offline. For this we used the Pinata API to pin the credential files on a distributed IPFS service, which slightly centralizes but keeps the files consistent. In a production deployment, each participating institution could run its own IPFS node to provide redundancy and reduce third-party pinning services.

5.4 Front-End Development

To demonstrate system usability a prototype web interface was built. Issuer Dashboard: Choose a student and credential type then upload the file using this simple form. When submitted, encryption is done and a MetaMask - triggered blockchain transaction occurs. This interface verifies issuer ID, displays unique credential ID and generates QR code for verification page. Student Dashboard: Students log in (currently via MetaMask accounts), see a list of issued credentials for their address and decrypt them for use personally. Sharing may involve creating verification links or QR codes. Future updates will allow integration with institution login systems. Verification Page: Get records directly from chain by entering credential ID or upload file to compute and check hash. The next page shows status (valid/revoked), issuer name and issuance date with visual indicators (green check for valid, red X for revoked). UX/UI testing showed the system takes little training. Students decrypted dummy credentials without developer help. Now styling is simple but design communicates system functionality. Private key management, transaction progress indicators & integrated QR code scanning are planned. Prototype web interface that shows how the system works for different users.

Issuer Dashboard: The issuer selects a student, a credential type and uploads the credential file via a simple form. On submission, the file is encrypted and a blockchain transaction is initiated through MetaMask. The

interface verifies issuer identity, shows newly generated credential ID and generates QR code (and link) for verification.

The screenshot displays the 'Certificate Validation System' interface. On the left, under 'Select Your Role', there are two options: 'Institute' (represented by the Victoria University logo) and 'Verifier' (represented by a building icon). On the right, the 'Generate Certificate' form is filled out with the following details:

Field	Value
UID	s8120994
Name	Bigyan Paudel
Course Name	Masters in IT (NMIT)
Org Name	Victoria University

A green success message states: 'Certificate successfully generated with ID: dd8755779139c5f8a859a74f48830d86998617e31e29ed2b710739f93ed7933c'. Below this, a link 'View Certificate on IPFS' is provided, followed by two debug logs: 'DEBUG: Looking up certificate ID: dd8755779139c5f8a859a74f48830d86998617e31e29ed2b710739f93ed7933c' and 'DEBUG: isVerified result: True'.

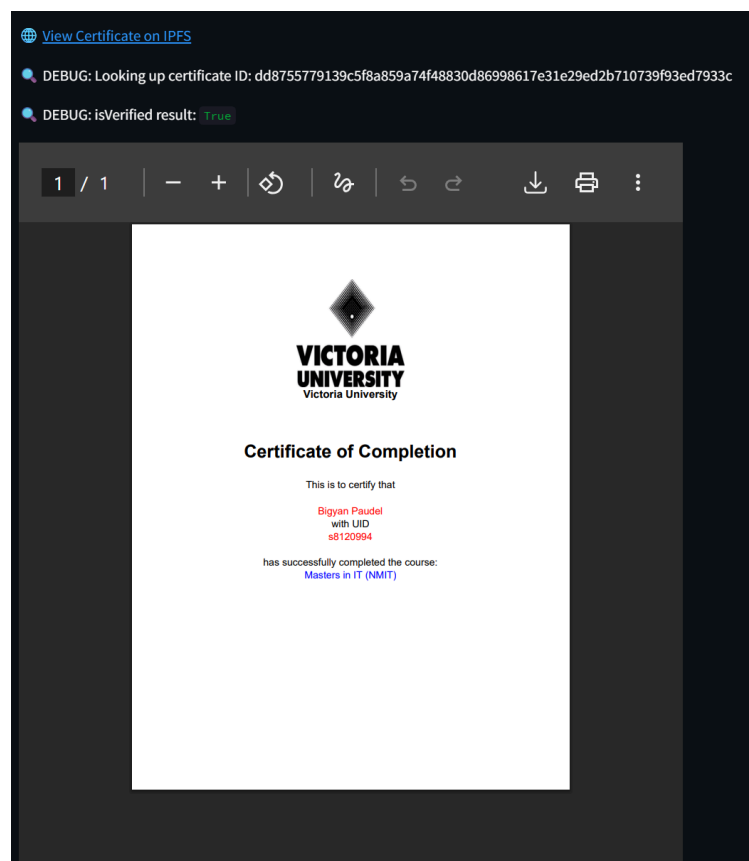


Figure 4,5,6: Certificate Issuance

Verification Page: A verifier enters a credential ID or uploads a credential file. The system calculates the hash of the file if one is provided and checks the blockchain. Then it shows the credential status (Valid or revoked), the issuing institution name and issuance date with visual indicators.

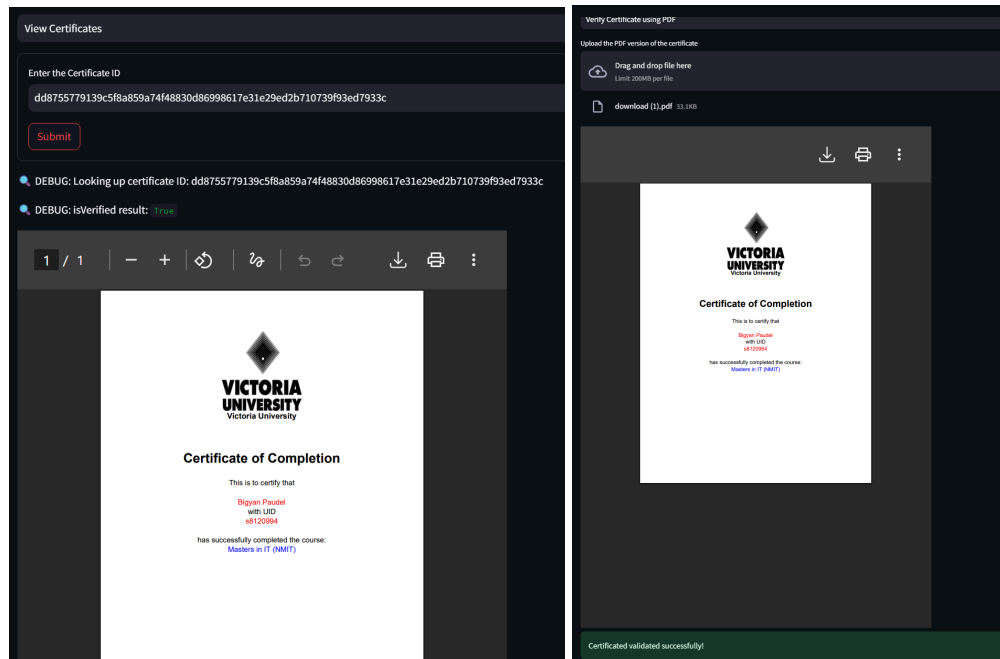


Figure 7,8: Verification Page

Initial usability testing showed little training is required. Trial runs showed students finding and decrypting test credentials without developer help. The prototype's interface is still basic in style but clearly communicates its intended functionality. Future work includes private key management UI improvements, transaction progress indicators and built in QR code scanning.

5.5 Testing & Progress of Results.

The prototype passed functional, performance, security and interoperability testing with good results in all categories.

Functional: End-to-end flows from issuing to revocation were executed with 10 test credentials. Warnings about revoked credentials appeared correctly on the verification page and student dashboard. Functional testing: We ran several end-to-end scenarios of issuing and revoking credentials (with ten test credentials in total). The system worked as expected; For example, a credential was revoked and the verification page and student dashboard showed it as revoked.

Performance: Issuing and encrypting took 1.5 seconds, blockchain confirmations took 3 seconds and retrieval/decryption 1 second. High Ethereum mainnet block times of 15 seconds still permit reasonable throughput and consortium chains could handle thousands of daily issuances. Performance testing: We found issuing a credential including file encryption took about 1.5 seconds, blockchain confirmation took around 3 seconds and credential file retrieval/decryption took about 1 second. With longer block times (15 seconds) on the public Ethereum network, the system throughput would remain acceptable. The platform could process on the order of thousands of credential issuances per day in a consortium network environment.

```
eth_call
eth_call

eth_call
eth_accounts
eth_getBlockByNumber
eth_estimateGas
eth_blockNumber
eth_getBlockByNumber
eth_sendTransaction

Transaction:
0x829d5c8b143918294ff312d0fc46ae490806a423c47de560d5044406f
3600edb
Gas usage: 181342
Block Number: 6
Block Time: Thu Sep 11 2025 21:03:15 GMT+0000
(Coordinated Universal Time)
```

Figure 9: Blockchain transaction execution

Security: Smart contract logic stopped Unauthorized issuance attempts. Credential tampering was prevented by blockchain immutability. Front end verification pulled live data directly from the chain so users were not misled by manipulated web content. Security testing: This prevented unauthorized credential issuance attempts and the blockchain was immutable so no issued credential could be altered or forged. Also, live data is always retrieved from the blockchain via the verification interface - no user can be duped by faked content on the web front-end.

Interoperability: Sample credentials were formatted in W3C Verifiable Credential JSON, not validated against standard verifiers due to custom networks but mapping confirmed compatibility potential with broader ecosystems. Interoperability testing: Sample Credentials were formatted according to the W3C Verifiable credentials JSON schema. They can not be fully validated with standard external verifiers as our tests were performed on a custom network, but this exercise showed that our credential format is potentially compatible with wider verification ecosystems.

These results show the prototype is functional, secure and efficient in comparison with traditional systems but more scalability and usability refinements are required. Overall the prototype is functional, secure and more efficient than traditional credentialing methods but still requires work on scalability and usability.

5.6 Work Completed versus Work Remaining.

Completed:

- Smart contract for credential lifecycle (issue, revoke & ver).
- setup of Multi node private blockchain with PoA consensus.
- IPFS, hybrid encryption fusion
- Dashboards for issuers, students & verifiers
- Core flows & interoperability checks passed Testing.

The main parts of the system are implemented so far. We have developed the smart contract for credential lifecycle including issuance, revocation and verification. A multi-node private blockchain network with PoA consensus mechanism is running. Off-chain storage was handled by IPFS combined with the hybrid encryption scheme. Simple web dashboards for issuers, students and verifiers to interact with the system were built. We have also tested the core end-to-end flows and interoperability checks - all initial tests were successful.

Remaining:

- Provide proper user authentication beyond wallet-only login.
- Optimize contracts for & search efficiency with lower gas usage.
- Test large file handling (multi-MB transcripts) & multi-institution scenarios.
- Improved UI/UX management & progress indicators & QR scanning with better key management.
- Comprehensive edge case testing (e. Pilot deployment documentation, user guides and stakeholder evaluations.)

A few tasks are planned to improve the system further. We need to implement proper user authentication instead of wallet login. The smart contracts can be optimized to cut gas costs and increase query efficiency. Tests with larger files such as multi-megabyte transcripts and simulations involving several institutions are planned. UX needs enhancements with improved user experience features, key management, clearer transaction progress indicators and integrated QR code scanning. Then we will do edge-case testing and preparations for a pilot deployment including user documentation, guides and stakeholder feedback during trial use.

The implementation phase has built a robust working prototype of a blockchain-based credential verification platform. Combining the Solidity smart contracts with IPFS (encrypted) and a functional web interface demonstrates secure, decentralized credential management is possible. Tests of functionality and performance show issuing, verification and revocation work as expected and much faster than with conventional systems. Those remaining tasks are to make the system usable, scalable and interoperable and ready for real-life trials. In all, the obtained progress contributes to the research problem resolution and to a transparent and tamper-proof credential ecosystem.

6. Discussion / Results

A working prototype of the blockchain-based academic credential verification is developed. Here we have addressed gaps identified from the literature namely lack of transparency, dependence on central databases and low student control of records. Prototype for blockchain-based academic credential verification is developed. Here we have addressed gaps identified from the literature namely lack of transparency, dependence on central databases and low student control of records. A functional prototype of the TrustEdu system has been developed which addresses some of the challenges described in the literature namely transparency, reliance on central databases & limited student control.

6.1. Prototype Development

Design and deploy contracts to register students, authorize issuers & create certificates. Each credential is stored with course title, completion date and issuer identity metadata on-chain. Issued certificates are tamper resistant because they are issued on blockchain. Any verifier can confirm authenticity without using one institution's database. Unlike other structures under which institutions own academic records, this framework grants students ownership. Their qualification is openly shared and proved directly. The

prototype uses smart contracts to register authorized issuers and to publish credential records on the blockchain.

Key metadata include course title, completion date and issuer identity. All certificates are tamper-evident because they are anchored on the blockchain, and any verifier can confirm a credential is genuine without relying on a single institution's database. The model also puts control over the individual - unlike traditional systems where institutions own records - students in TrustEdu own their credentials and can share them directly when needed.

6.2. Comparison with Existing Solutions

Our prototype was compared with popular platforms where following innovations were highlighted: We contextualized TrustEdu against three existing digital credential platforms - My eQuals, Digitary & MIT's Blockcerts. Table 2 summarizes that comparison.

Criteria	TrustEdu (Our Platform)	My eQuals	Digitary	Blockcerts (MIT)
Decentralization	Permissioned blockchain across institutions; no single point of failure	Centralized, single platform	Centralized vendor network	Public blockchain anchoring, no central repo
User Control	High – students hold cryptographic keys and manage credentials	Low – platform controls sharing	Low – platform operator controls sharing	High – student-owned wallet, no reliance on issuer
Revocation	On-chain registry; verifiers automatically see revoked status	Revocation handled centrally, offline PDFs may not update	Managed in network only, not external	No standardized revocation method
Interoperability	Designed for open standards (W3C VC, DID)	Limited, mostly national/regional	Limited to Digitary ecosystems	Potentially interoperable, but requires external tools
Privacy	Sensitive data encrypted off-chain; only hashes stored on-chain	Moderate – central storage	Moderate – central storage	Moderate – student controls file, but full data visible when shared

Table 2: TrustEdu against existing digital credential platforms

In Table 2, TrustEdu differentiates itself by removing single points of failure via decentralization and by giving users direct control of their credentials. Currently, platforms like My eQuals and Digitary all use centralized infrastructure and control whereas Blockcerts was the first such platform to use such centralized infrastructure and control.

It supports decentralization and user ownership but not revocation by itself. TrustEdu combines those benefits - decentralized trust, robust user ownership & integrated revocation.

6.3. Literature Gap Addressed.

Reputation replaced verifiable technology. Blockchain removes this dependency by adding trust into the protocol itself. Most credentialing systems store credentials in isolated systems that make international verification difficult. Blockchain enables a common standard for issuing and verifying across borders. Earlier systems allowed less student agency. Our model values individual control so learners own their success. This prototype addresses the trust deficit in the current systems by using blockchain as a trust mechanism instead of institutional reputation. It also improves interoperability by combining credential verification across institutions and countries. Thirdly, it is designed for student-centric ownership of credentials, giving learners control of their academic records in ways that traditional systems could not.

6.4. Early Testing & Validation

Core contract tasks include creation of students, registration of issuers & issuing certificates have been mock tested. This system receives certificates returned and verified by smart contract queries. First feedback suggests the approach might cut administrative overhead and build confidence for employers, universities and professional bodies. Preliminary functional tests such as creating students, registering issuers and issuing certificates were conducted and the core contract functions are working correctly. Verification was also validated as the system got certificates verified through smart contract queries. Initial user feedback suggests our approach could reduce administrative burden and increase credential authenticity confidence for employers and academic institutions.

6.5. Contribution to Research Objectives

Failed to combat fraud and increase transparency contributed a working implementation that complements theoretical frameworks considered in academic research. Build upon a solution that can be compared/benchmarked against real world platforms. This project tackles academic credential fraud directly through transparent verification. It is a working implementation supplementing theoretical frameworks from prior research, bridging the gap between theory and practice. Also, it provides the basis for an eventual solution to be benchmarked against real world platforms in future work.

7. Future Work

Though promising, more work remains to refine, scale and prepare the system for real world adoption.

7.1. Technical Enhancements

- Scalability: Streamline the smart contract design to reduce gas costs and optimize performance, especially for large volumes of credentials.
- Data Privacy: Add selective disclosure and off-chain storage to meet privacy regulations like GDPR without compromising verifiability.
- Interoperability: Develop APIs so existing student information systems in universities can communicate with the blockchain.

7.2. Security & Compliance

- Encryption: Strengthen student data protection improved.
- Regulatory Alignment: Compatibility with global frameworks like GDPR, Australian Privacy Principles and ISO standards for digital records.
- Governance: Consider models where consortiums of universities maintain the network while balancing decentralization with accountability.

7.3. User-Focused Improvements

- Interface Design: Create a portal for students, issuers and verifiers.
- Selective Sharing: Permit students to share only relevant data with employers (e.g. degree title but not grades).
- Accessibility: Seek to make the system work across devices & accessible to all learners - even those with limited technical skills.

7.4. Comparative Benchmarking

Systematic comparisons with existing platforms:

- MyEquals: Measure improvements in decentralization, speed and cross border recognition.
- Digitary: Assess gains in interoperability and transparency.
- Emerging Blockchain Pilots: Demonstrating how this system can deliver better governance and global usability.

7.5. Pilot Deployment

- Work with a partner institution to pilot the framework in a controlled environment.
- Gather feedback from students, academic staff and employers on requirements.
- Measuring outcomes like verification speed, fraud detection and administrative cost savings.

7.6. Long-Term Vision

Long term, we see TrustEdu as an academic and professional credentials marketplace connecting educational institutions, employers and government agencies. This could be extended to include micro-credentials, online course certificates and professional licenses besides university degrees. We hope to eventually help define new global standards for secure and transparent blockchain-based digital credentialing.

Work done to date points towards a blockchain-based academic credential verification system that addresses weaknesses of centralized platforms. Through smart contract implementation, IPFS integration and initial front-end development the prototype demonstrates that a decentralized model can offer secure issuance, tamper-proof storage and verifiable sharing of academic records. That is far better than current digital credential services like MyEquals and Digitary which although functional are tied to centralized infrastructures that often impede student autonomy and transparency. Its greatest contribution is emphasis on student ownership of data. With files encrypted with student-controlled keys and storage decentralized via IPFS, learners remain the administrators of their academic achievements - not institutions. And this shift empowers students and builds trust with outside verifiers who can rely on blockchain-backed authenticity without having to contact the issuing institution directly. With increasing academic mobility and cross-border verification this approach would ensure more efficiency and fairness in the recognition of qualifications. The underlying architecture is technically viable as demonstrated by the deployment of Solidity smart contracts and early testing of credential issuance, revocation and verification flows. Tested transaction times ,

encryption speeds, integration tests show that the solution can handle real workloads even at a prototype stage. It demonstrates that in a permissioned or consortium setting, blockchain can deliver theoretical benefits alongside operational efficiency. The fact that revocation and verification were validated by functional tests supports the reliability argument in real usage. Work done to date points towards a blockchain-based academic credential verification system that addresses weaknesses of centralized platforms. The prototype shows through smart contract implementation, IPFS integration and initial front-end development that a decentralized model can offer secure issuance, tamper-proof storage and verifiable sharing of academic records. That is far better than current digital credential services like MyEquals and Digitary which although functional are tied to centralized infrastructures that often impede student autonomy and transparency. Its greatest contribution is emphasis on student ownership of data. With files encrypted with student-controlled keys and storage decentralized via IPFS, learners remain the administrators of their academic achievements - not institutions. This shift empowers students and builds trust with external verifiers who can rely on blockchain-backed authenticity without contacting the issuing institution directly. In a world of increasing academic mobility and cross-border verification this approach allows more fair recognition of qualifications. The underlying architecture is technically viable as demonstrated by the deployment of Solidity smart contracts and early testing of credential issuance, revocation and verification flows. Tested transaction times, encryption speeds and integration tests show that even the prototype can handle real workloads. This demonstrates that in a permissioned or consortium setting, blockchain can deliver theoretical benefits alongside real operational efficiency. Revocation and verification through functional tests are successfully validated as reliable in real use.

But even its foundation remains to be refined. Moving from RSA to elliptic curve cryptography or more sophisticated key management solutions will be required to scale adoption. Regulations like GDPR compliance also have to be considered during design before institutional rollouts. Also, while IPFS allows decentralization benefits, files will require long-term persistence under governance; Multi-institutional pinning strategies will be needed to achieve reliability without compromising decentralization. The broader research contribution is to fill gaps in the literature on interoperability and trust. Aligning the prototype with emerging standards like W3C Verifiable Credentials this project shows that blockchain-backed academic records need not be isolated. But even its foundation remains to be refined. Upgrades to security - moving from RSA to elliptic curve cryptography or more sophisticated key management solutions - will be necessary as we scale up. Regulations like GDPR have to be considered too before any institutional rollouts. Also, although IPFS provides decentralization benefits, file persistence over a long term under a proper governance model will be required - multi-institutional pinning strategies will be needed to maintain file availability without compromising decentralization. Its wider contribution is to close interoperability and trust gaps that remain. This project demonstrates that blockchain-backed academic records do not have to be isolated - aligning the prototype with emerging standards like W3C Verifiable Credentials.

Employers and professional bodies could validate a credential instantly without lengthy manual checks reducing costs and risk of fraudulent qualifications. And students get more: they receive lifetime, portable, secure credentials - regardless of whether the issuing institution remains open. Future development phases, pilot rollout with select institutions, comprehensive stress testing and comparative benchmarking against centralized solutions will provide the evidence base for large scale adoption. It may also change the way academic credentials are issued and trusted worldwide, setting a new standard for transparency, efficiency and equity in education systems if it succeeds. Employers and professional bodies could validate a credential instantly without lengthy manual checks reducing costs and risk of fraudulent qualifications. And students get more: they get lifetime, portable and secure credentials whether the issuing institution exists or not. Future planned large-scale adoption evidence will come from pilot rollouts with select institutions, comprehensive stress testing and comparative benchmarking against centralized solutions. It may also change how academic

credentials are issued and trusted worldwide, setting a new standard for transparency, efficiency and equity in education systems if it succeeds.

Conclusion

This research is more than a technical prototype made out of paper: it is a philosophical research. Trace the evolution of a new paradigm for academic credential verification. By combining blockchain, smart contracts, decentralized storage and encryption the work demonstrates that a secure and scalable alternative to centralized credential platforms is possible and credible. Its TrustEdu prototype shows how students can own their academic achievements in a transparent but privacy-preserving manner and how institutions can issue verifiable credentials without intermediaries.

Beyond the technical contribution the project marks a social change. It envisages a future in which learners are no longer passive recipients of institutional records but active custodians of their own academic history. So this empowerment builds individual agency while promoting mobility across borders as students can share their credentials with universities, employers and government agencies worldwide. In parallel, institutions are urged to operate in a decentralized trust ecosystem free of centralized authorities and open to global collaboration.

The implications of this research thus go beyond a proof-of-concept. This model challenges the current model of credential management - one that is decentralized, interoperable and therefore resistant to fraud. This likewise aligns with wider movements around digital identity and data sovereignty and a vision of a higher education where transparency and trust are built into the infrastructure itself.

In summary, this research is not about the success of a prototype but about the recognition that blockchain-enabled credential verification can be the foundation of a student-centric and globally interoperable framework. Work in credentialing advances a technological vision while also offering a social vision: a future where learners worldwide own their academic achievements & institutions collectively uphold trust in education through decentralized collaboration.

References

- Ambast, S. K., & Sumesh, T. A. (2022). A blockchain based credential verification system using IPFS. *Proceedings of the 2022 IEEE 19th India Council International Conference (INDICON)*, 1–5. IEEE.
- Borenovic Dilas, D. (2021). Digital Academic Records: A Credential Evaluator's Perspective. *World Education News & Reviews (WENR)*, June 22, 2021.
- Kabashi, F., Snopce, H., Luma, A., & Neziri, V. (2024). Trustworthy Verification of Academic Credentials through Blockchain Technology. *International Journal of Online and Biomedical Engineering*, 20(09), 51–64.
- Li, Z. Z., Liu, J. K., Yu, J., & Gašević, D. (2022). Blockchain-based Solutions for Education Credentialing System: Comparison and Implications for Future Development. *5th IEEE International Conference on Blockchain (ICBC)*, 266–273.
- Noshi, [N.], & Xu, Y. (2024). Development of blockchain-based academic credential verification system. *Open Access Library Journal*, 11, 1–19.
- OpenCerts. (2019). *OpenCerts – An open-source schema for issuing and verifying certificates on Ethereum* [White paper]. GovTech Singapore and OpenCerts Consortium.
- Pandya, S. (2022). Innovative blockchain solutions for enhanced security and verifiability of academic credentials. *International Journal of Science and Research Archive*, 6(1), 347–357.
- Patel, H. B., & Kaneriya, J. (2023). A secure and privacy-preserving student credential verification system using

blockchain technology. *International Journal of Information and Education Technology*, 13(8), 1251–1260.

- Rahman, T., Mouno, S. I., Raatul, A. M., Al Azad, A. K., & Mansoor, N. (2023). Verifi-Chain: A credentials verifier using blockchain and IPFS. *arXiv preprint arXiv:2307.05797* (presented at ICCCT 2023).
- Tariq, A., Binte Haq, H., & Ali, S. T. (2023). Cerberus: A blockchain-based accreditation and degree verification system. *IEEE Transactions on Computational Social Systems*, 10(4), 1503–1514.
- Wang, Y., & Cong, X. (2023). Blockchain for credibility in educational development: Key technology, application potential, and performance evaluation. *Security and Communication Networks*, 2023, Article ID 5614241.